No. of Functional Features Included in the Solution

Date	27 June 2026
Team ID	SWUID2026217448
Project Name	A Comprehensive Measure of Well-Being - HDI Predictor
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	HDI Score Prediction	Predict HDI score using 4 input indicators	ML Model	Done	High
2	HDI Category Classification	Classify as Very High / High / Medium / Low	ML Model	Done	High
3	Data Preprocessing	Handle null values, label encoding, feature selection	Data Processing	Done	High
4	Data Visualization	EDA with heatmaps, histograms, scatter plots	Visualization	Done	Medium
5	Linear Regression Training	Train model with 75/25 split, R2 = 97.73%	ML Model	Done	High
6	Model Saving	Save trained model using Pickle as HDI.pkl	Storage	Done	High
7	Flask Home Page	Input form with 4 HDI indicator fields	Web Application	Done	High
8	Prediction Result Page	Display HDI score and category to user	Web Application	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6

Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Home page form, Result page display	indexnew.html, resultnew.html
2	Backend / Logic	Linear Regression model, Flask routes, Pickle loader	app.py, HDI.pkl
3	Database / Storage	CSV dataset, Pickle model files	HDI.pkl, label_encoder.pkl, CSV
4	API / Integration	No external API integration used	Local Flask server only
5	Security / Authentication	No authentication required	Open access web application